

Aylin Davoudi

Toulouse, France | (+33) 647 716 687 | Ayvindavoudi2003@outlook.com | <https://linkedin.com/in/aylin-davoudi>

Website: <https://aylindavoudi.github.io/AylinDavoudi/>

PROFILE

Aerospace engineering master's student with a multidisciplinary background across diverse engineering projects, specializing in space propulsion. Seeking a 4-6 month summer end-of-studies internship starting in April/May 2027.

EDUCATION

Master of Aerospace Engineering (Space Systems Major) – ISAE-SUPAERO, Toulouse, France

Relevant Coursework: Space Propulsion, Space Systems, Satellite Thermal Control Systems, Computational & Experimental Fluid Dynamics, Satellite Propulsion: Chemical and Electrical, Space Environment and Space Law, Launchers Architecture, Acoustics 2025–2027

International Bachelor in Mechanical, Materials & Aerospace Engineering – INSA Lyon

Relevant Coursework: Heat Treatments, Material Science, Composite Materials, Machine Elements, Transducers 2022–2025

Courses at ECAM LaSalle (partnership with IBMMAE, INSA Lyon):

Compressible Flows & Propulsion, Thermodynamics, Manufacturing & CNC Machining 2022–2025

Exchange Semester – University of Twente, The Netherlands:

Fluid Dynamics, Heat Transfer, Control and Mechatronics FEB– JUN 2024

PROJECTS

Droplet Evaporation in Real-Gas & Supercritical Conditions – Master's Research Project ISAE-SUPAERO 2025–Ongoing

- Extended a Python-based D²-law/Abramzon–Sirignano evaporation solver to real-gas conditions using Peng–Robinson EOS, multi-component mixing rules, and fugacity-based thermodynamic corrections.
- Developing a diffuse-interface Langmuir–Knudsen treatment coupled with a Helmholtz free-energy EOS to extend the model to supercritical conditions.

Design & Optimization of a Pintle Injector – Bachelor Thesis INSA Lyon 2024–2025

- Designed and experimentally validated a RP-1/LOX pintle injector for lander propulsion using analytical mass-flow/momentum models and Star-CCM+ CFD to optimize pintle angle and gap for mixing and thrust.

Design & Optimization of Regenerative Cooling Channels for Vulcain 2 – University of Twente 2024

- Designed and optimized regenerative cooling channels for a Vulcain 2 rocket nozzle using analytical heat-transfer models and CFD, with wind-tunnel experiments validating predicted cooling performance.

WORK EXPERIENCE

Thermal Cooling Team Leader — PERSEUS (CNES) / ISAE SUPAERO 2026 – present

- Designing regenerative cooling circuits for 15 kN LOX–ethanol rocket engines.

Engineering Intern — CMD Gears, Cambrai, France Apr 2025 – Aug 2025

- Performed analytical and FEM parametric studies of conical interference fits, bearings, gears and lubrication, evaluating contact pressure, torque transmission and structural behavior using FEMAP, Solid Edge and KISSsoft/KISSsys.
- Delivered technical reports and engineering justifications for internal and client projects.

Assembly Line Worker — Stellantis, Poissy, France Jun – Aug 2023

- Performed assembly operations (kitting, leak testing, wheel and bumper assembly) on automotive production lines.
- Developed precision, adaptability, and teamwork in a fast-paced manufacturing environment.

SKILLS

Programming Python, C/C++, HTML, CSS, JavaScript, Fortran

Software MATLAB/Simulink, SolidWorks, Solid Edge, Fusion 360, Ansys, Star-CCM+, COMSOL Multiphysics, Abaqus, FEMAP, KISSsoft/KISSsys, XFOIL/XFLR5, OpenVSP, IBM Engineering Rhapsody (SysML), SU2, GetFEM++, Gmsh, ParaView, Microsoft Office, GitHub/GitLab ISAE softwares: Longisim, Transsim. Timeus

Machining Certified HAAS lathe and mill operator, G-code programming

Languages English (IELTS Academic 8.5), French (intermediate), Spanish (intermediate), Portuguese (beginner)